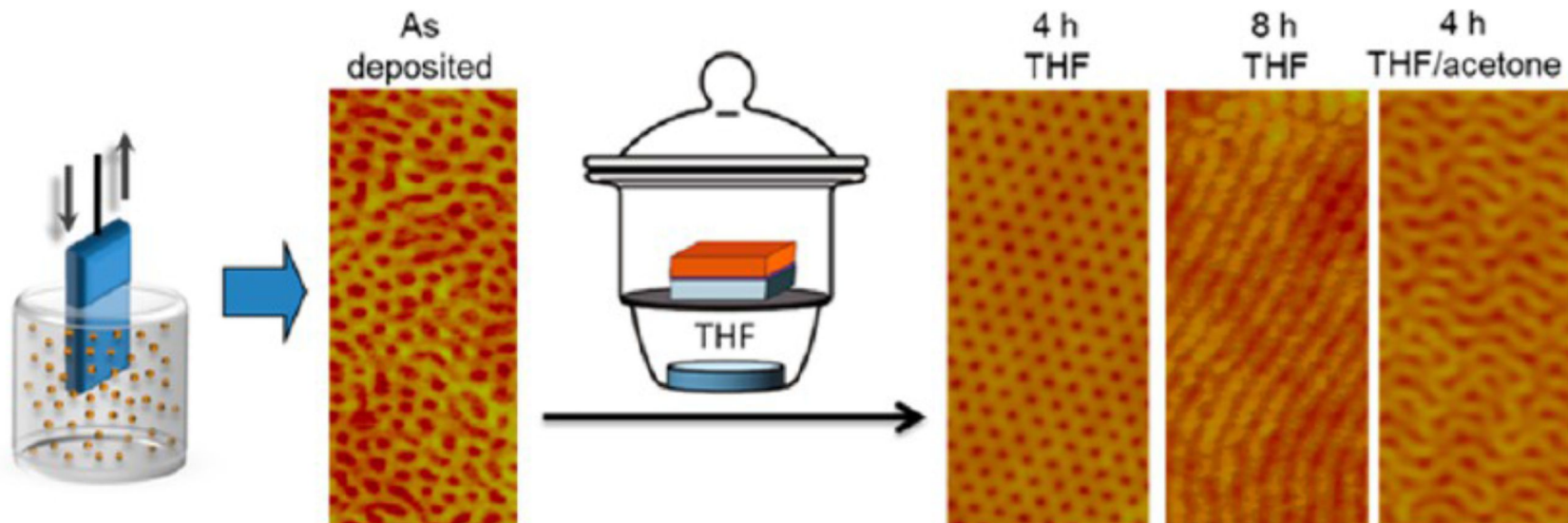


Two Important Challenges

- Long range order (regularity of the patterns)

Thermal annealing Place the sample in a vacuum oven at a temperature significantly higher than T_g of both blocks.

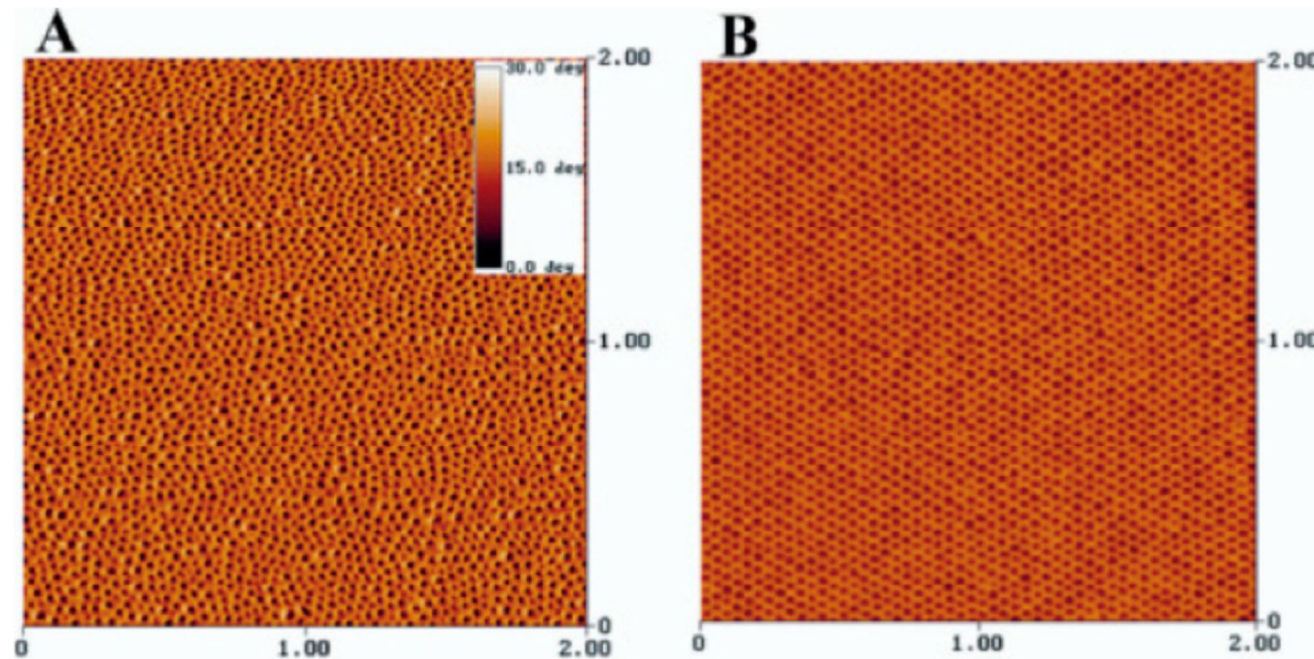
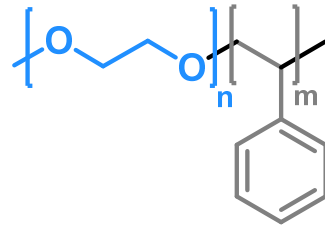
Solvent annealing



Quality of the Patterns – Long Range Order

Annealing

Solvent annealing

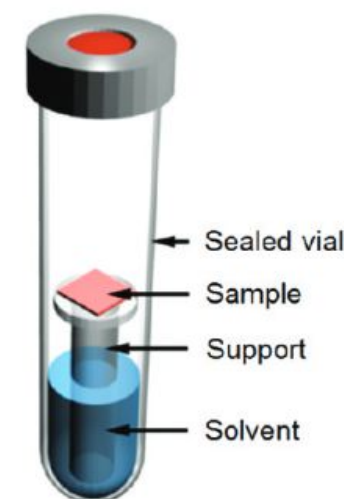
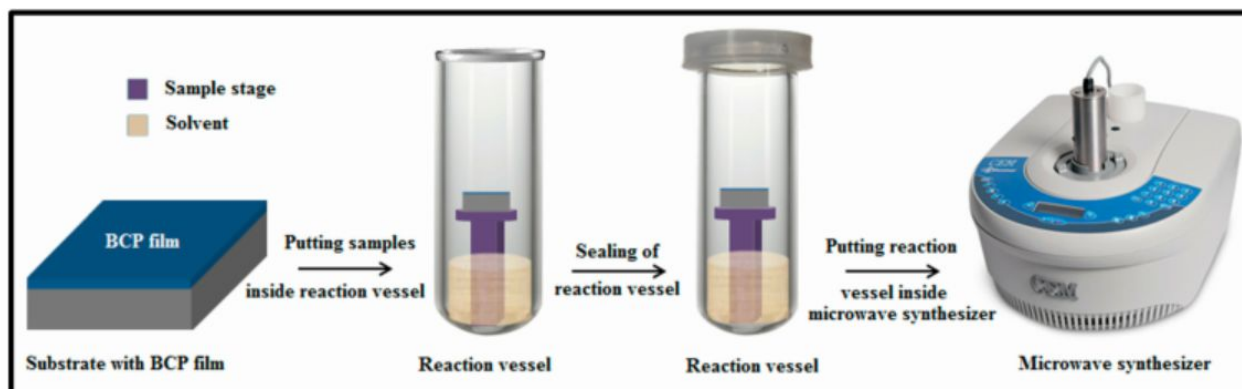


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Annealing

Solvent and thermal annealing

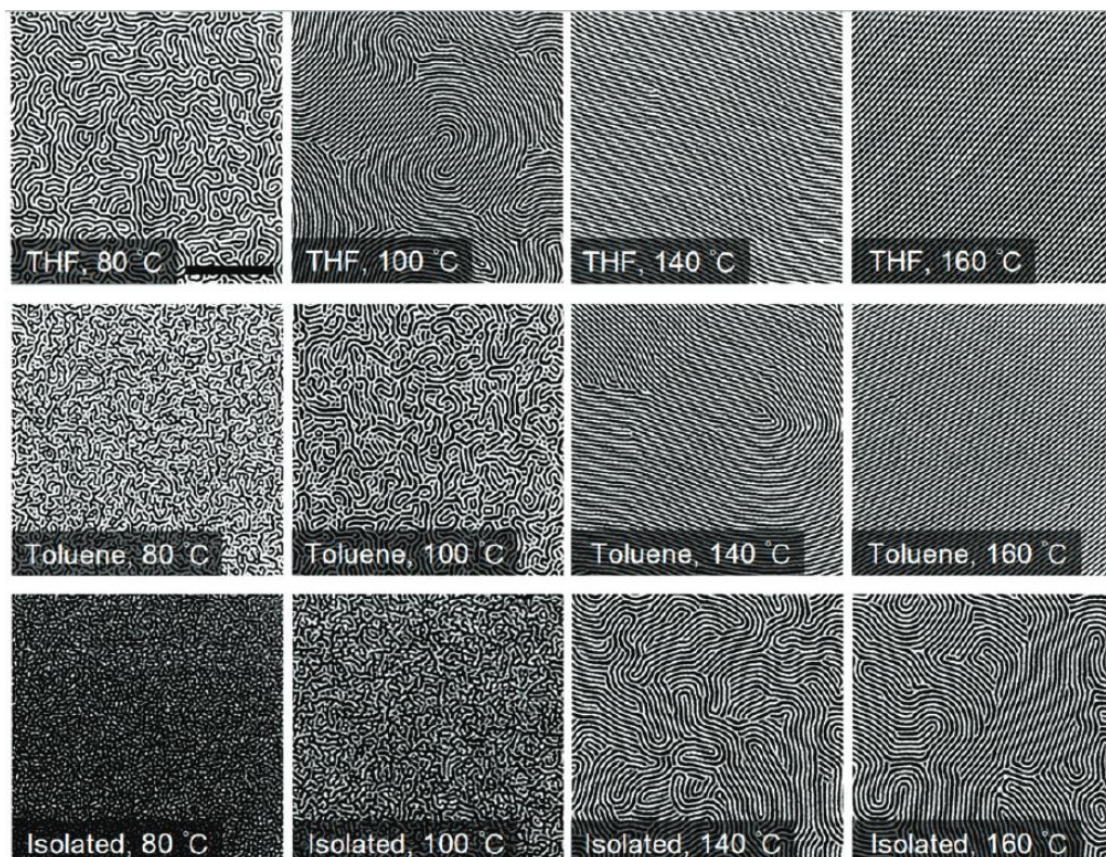
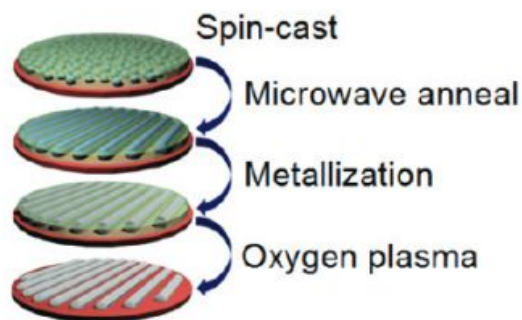
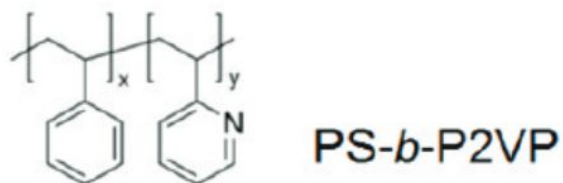


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Annealing

Solvent and thermal annealing



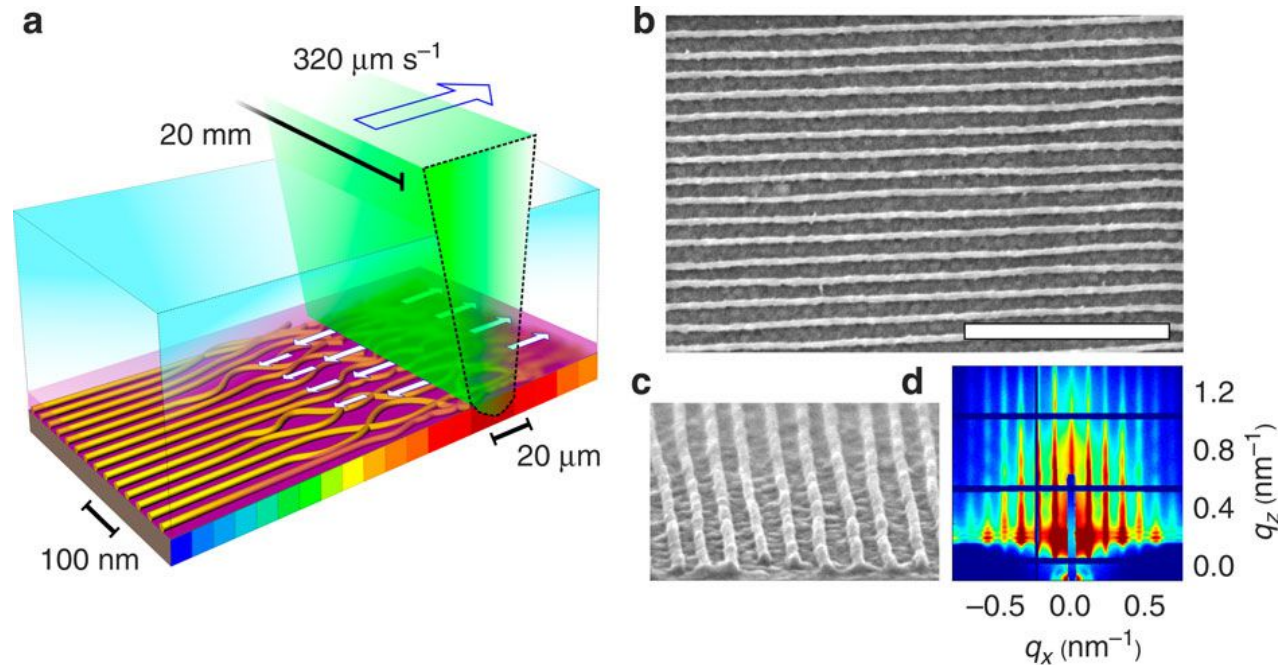
SEM of Pt structures obtained from annealed PS-*b*-P2VP films

Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Annealing + Directing

Thermal annealing

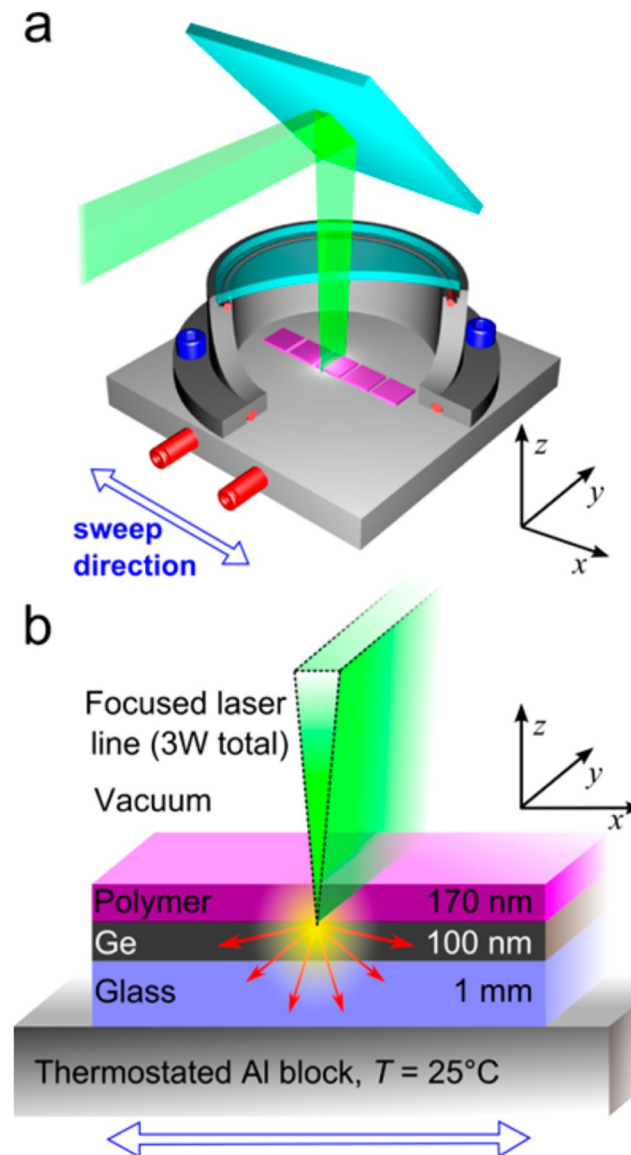


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Annealing + Directing

Thermal annealing

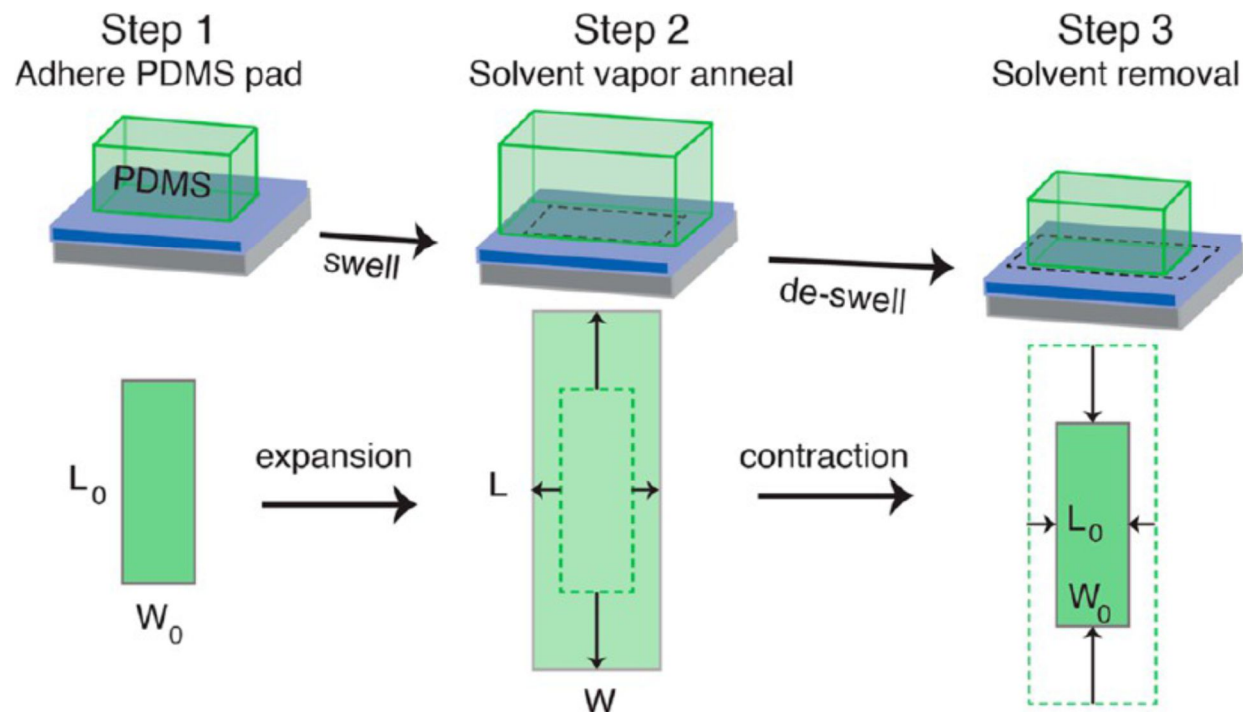


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Annealing + Directing

Solvent annealing

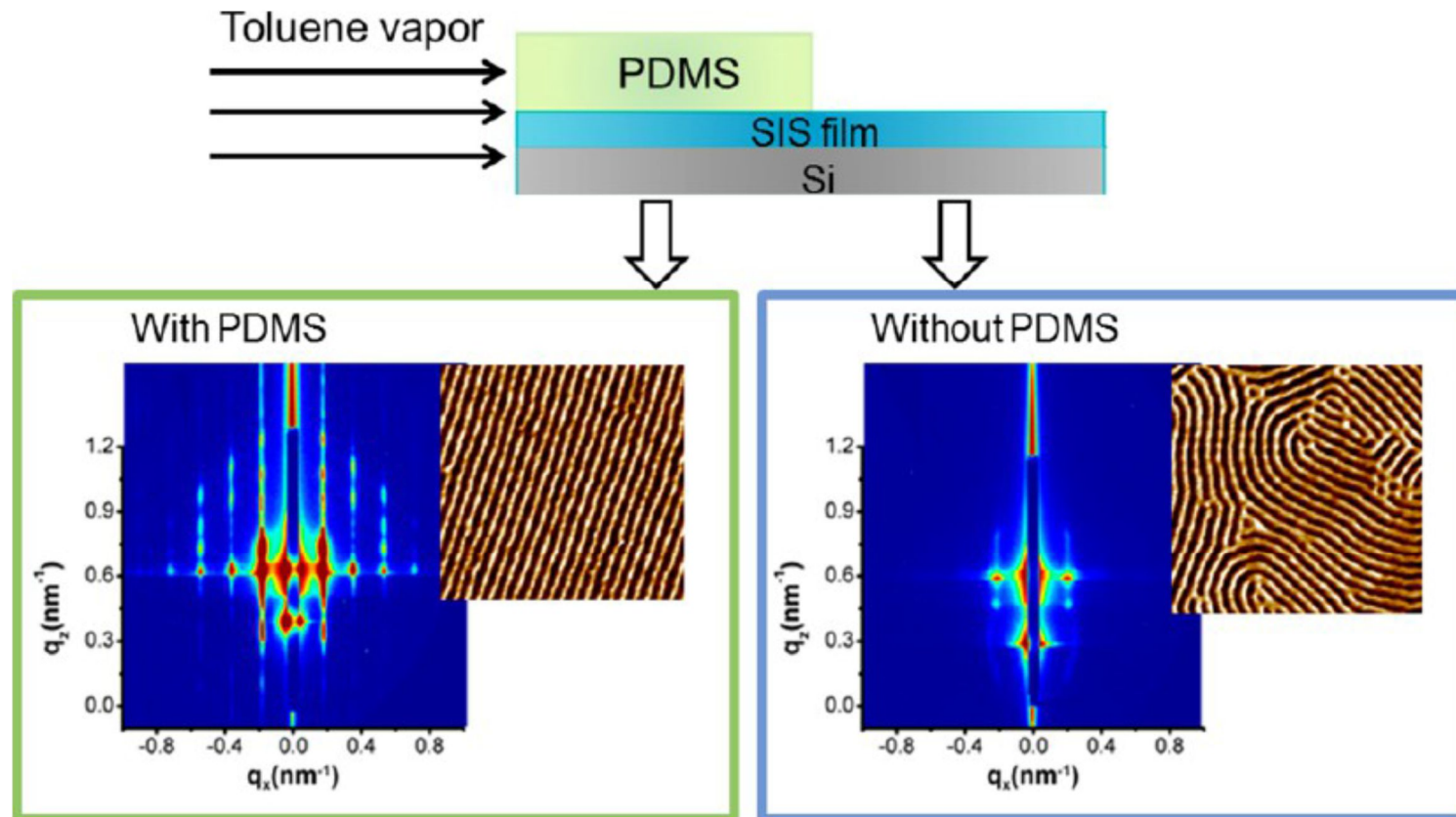


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Annealing + Directing

Solvent annealing

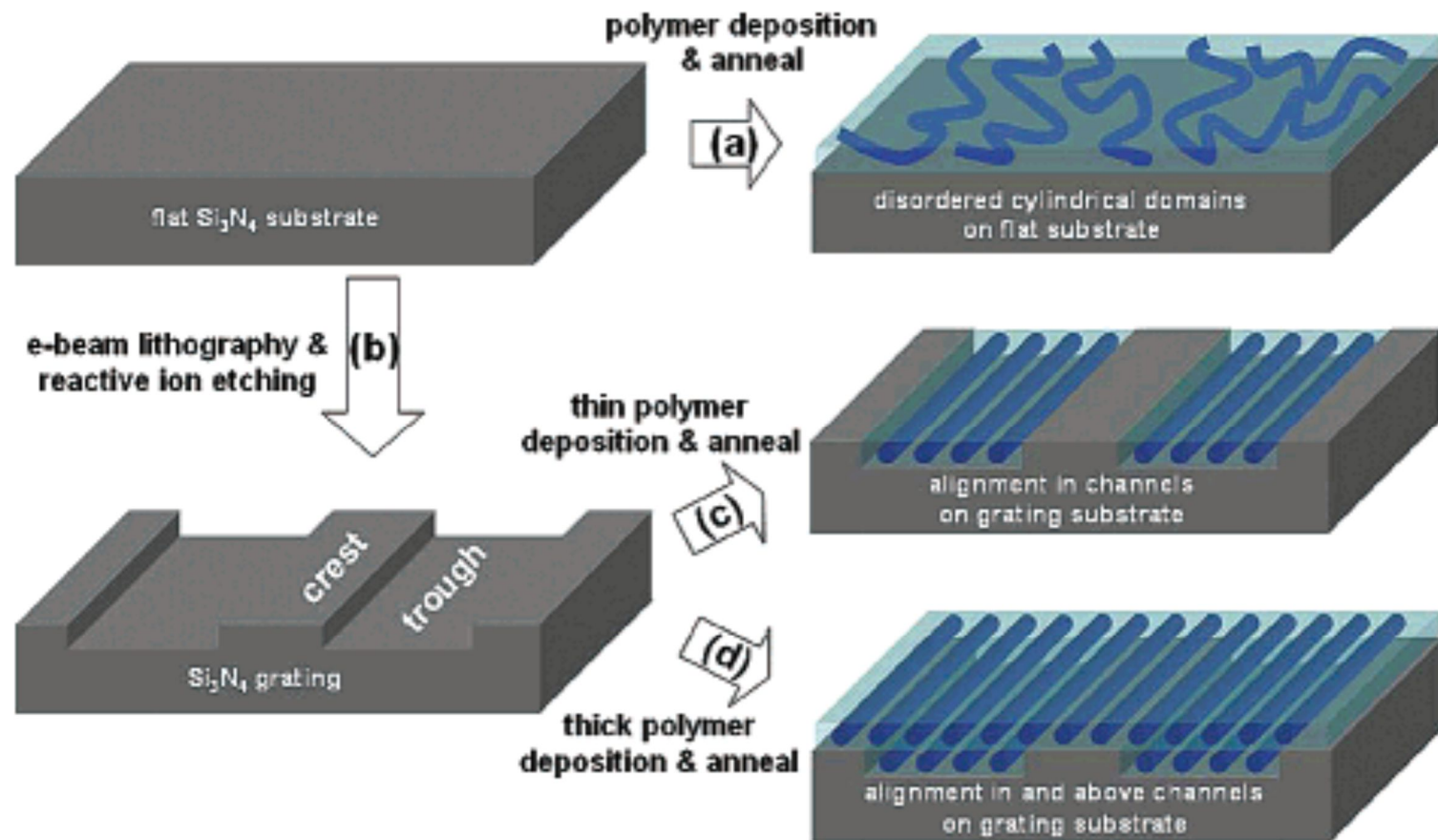


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Directed Self-Assembly

Topographical Patterns

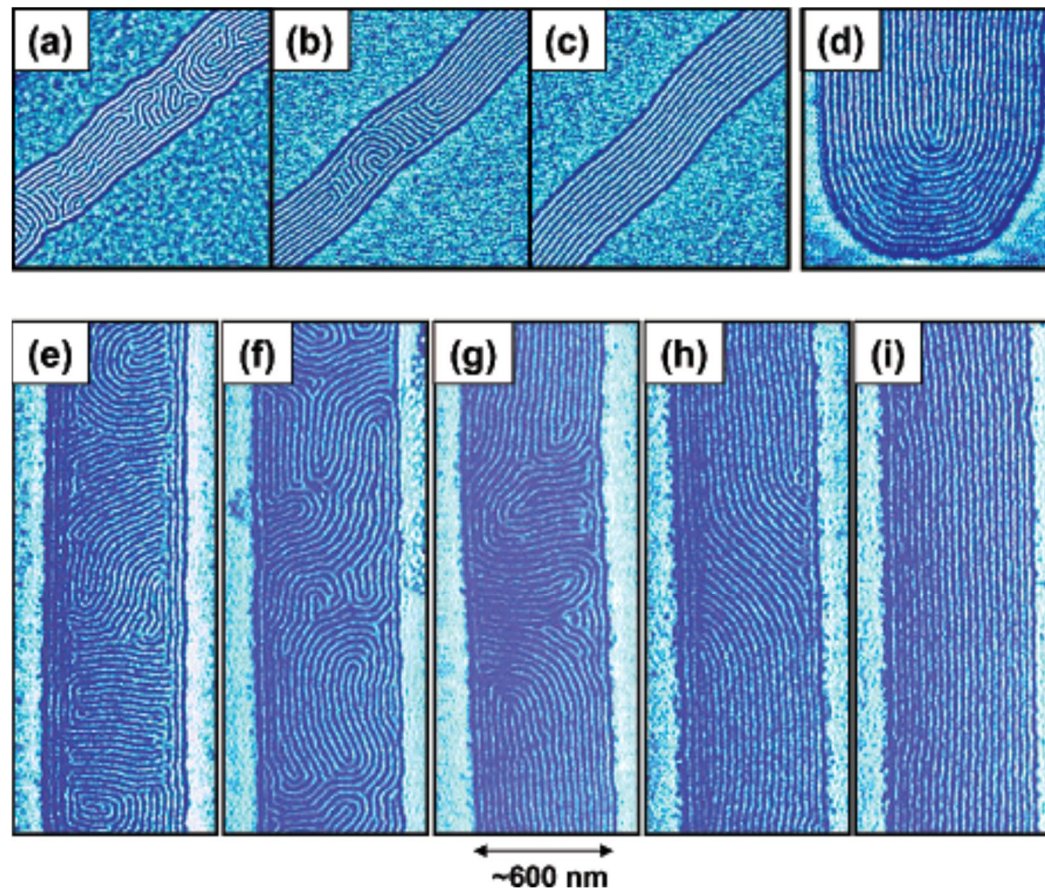


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Directed Self-Assembly

Topographical Patterns

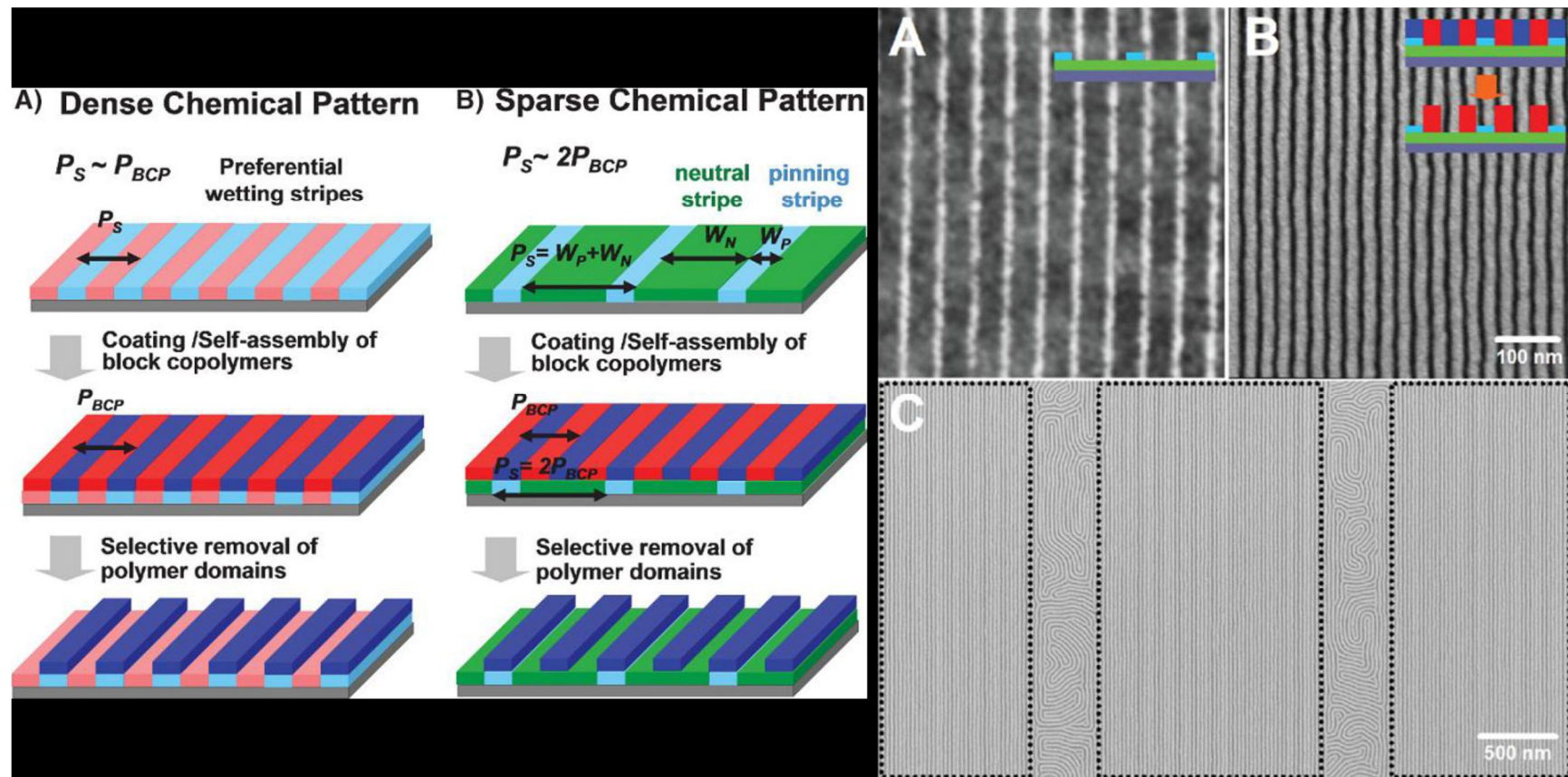


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Directed Self-Assembly

Chemical Patterns

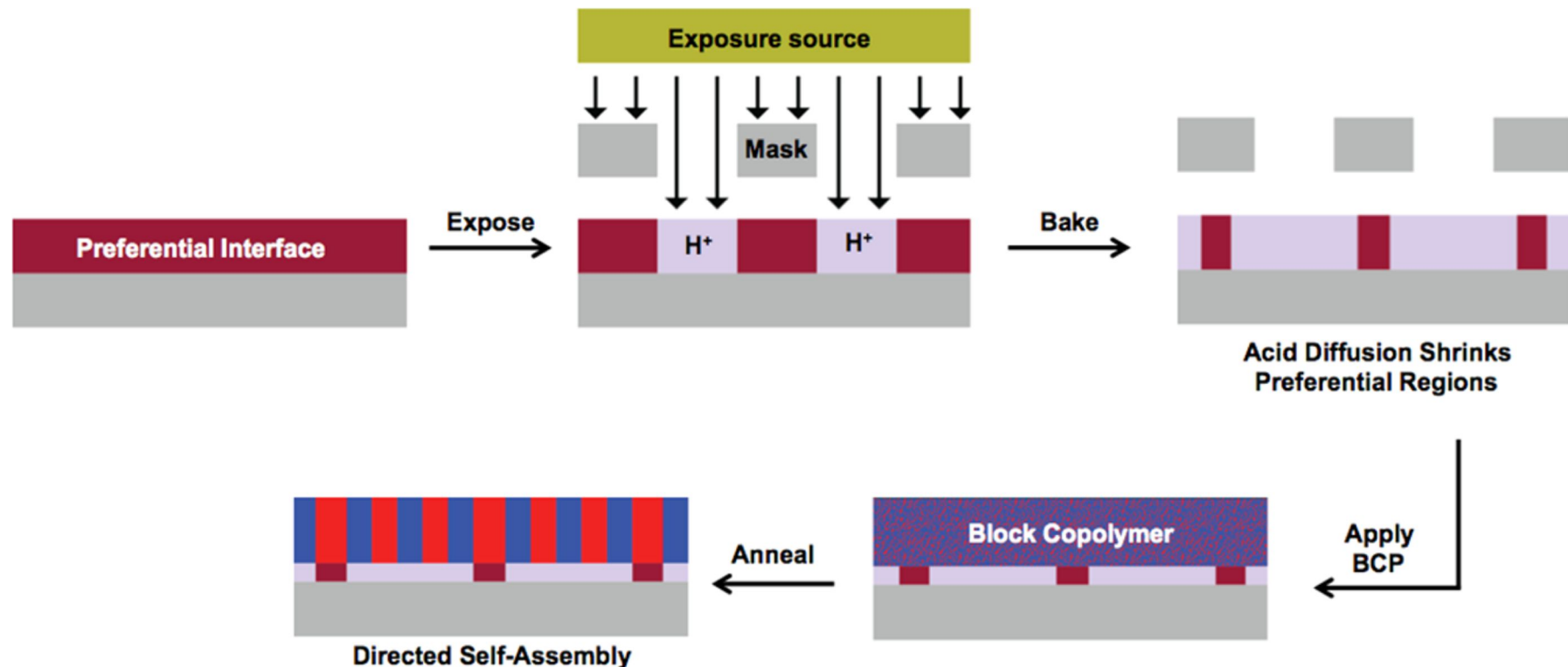


Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

Directed Self-Assembly

Chemical Patterns



Applications of Bulk Self-Assembly

Quality of the Patterns – Long Range Order

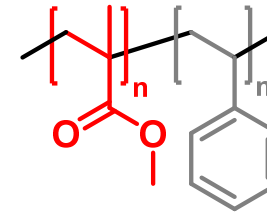
Directed Self-Assembly

Additional : Electric or magnetic fields

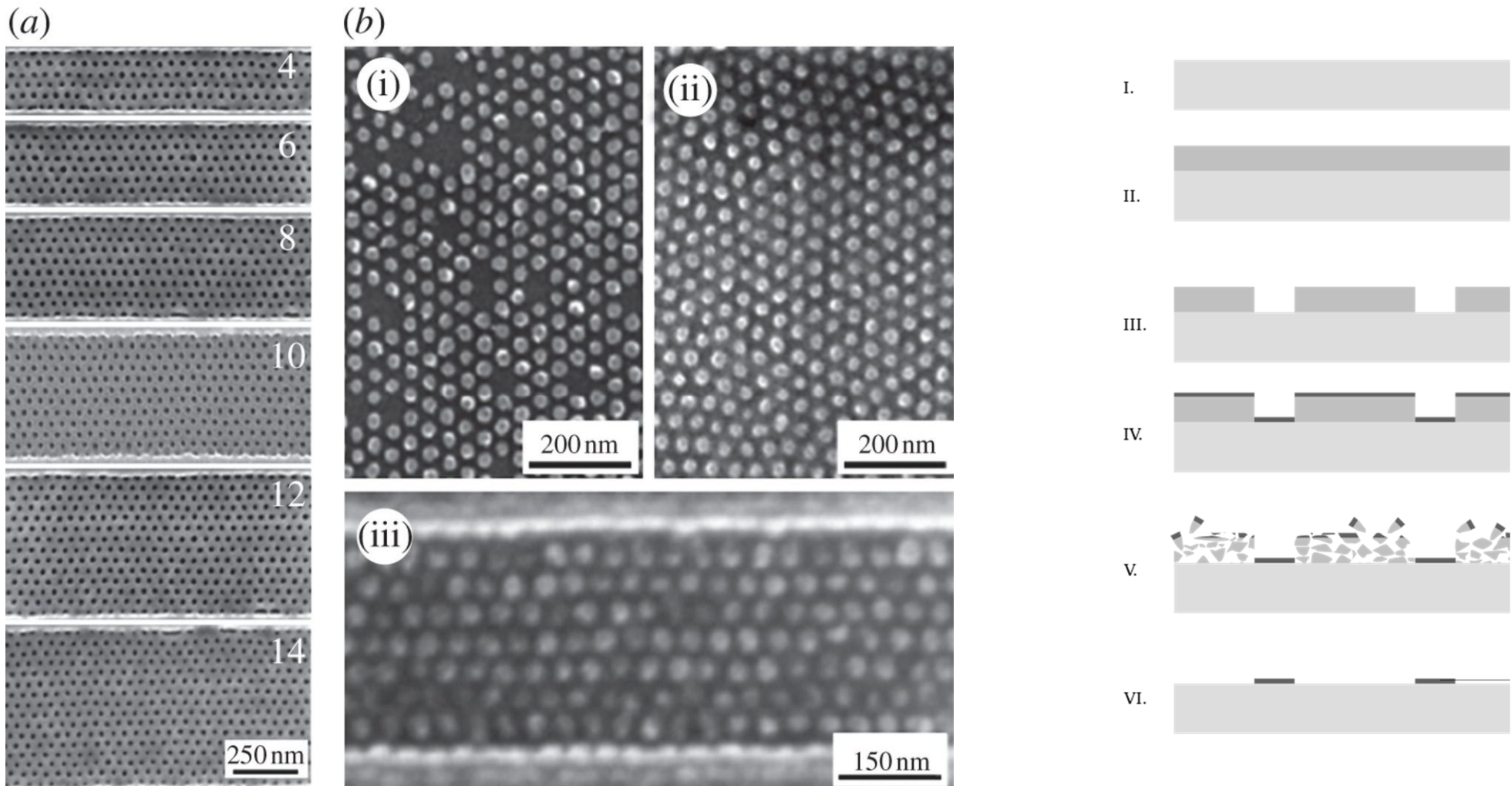
Applications of Bulk Self-Assembly

Selective Removal of One Domain

Templating for Materials Immobilization/Growth



Cobalt dots by evaporation and lift-off



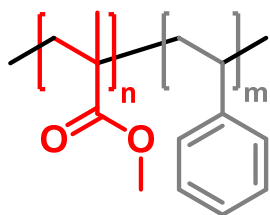
Nanotechnology 2005, 16, S324

Applications of Bulk Self-Assembly

Selective Removal of One Domain

Templating for Materials Immobilization/Growth

Sequential infiltration synthesis (SIS)

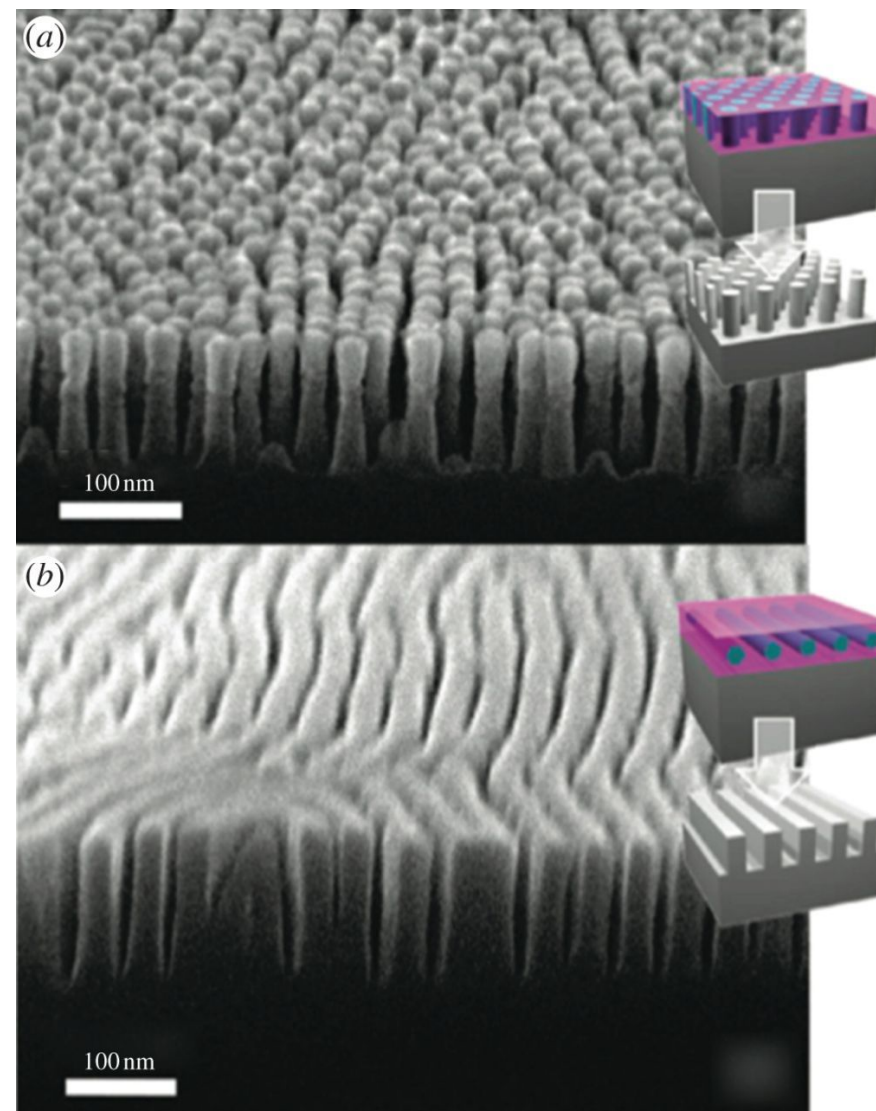


Atomic layer deposition of $\text{Al}(\text{CH}_3)_3$

Then non-selective oxygen RIE or heating to 500 °C

Could be done as well with TiO_2 , SiO_2 , ZnO , and W

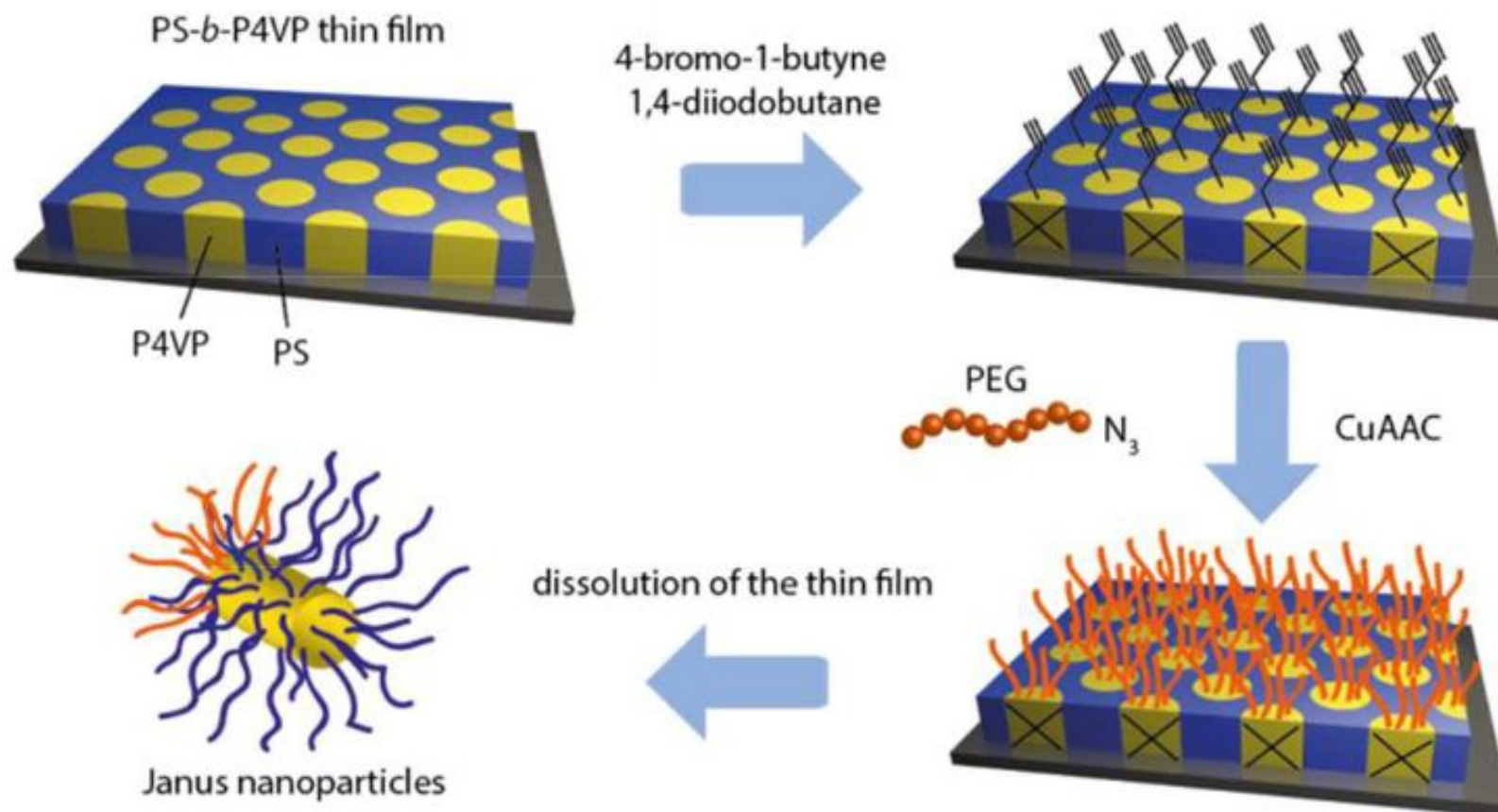
Adv. Mater. **2010**, 22, 5129
ACS Nano **2011**, 5, 4600

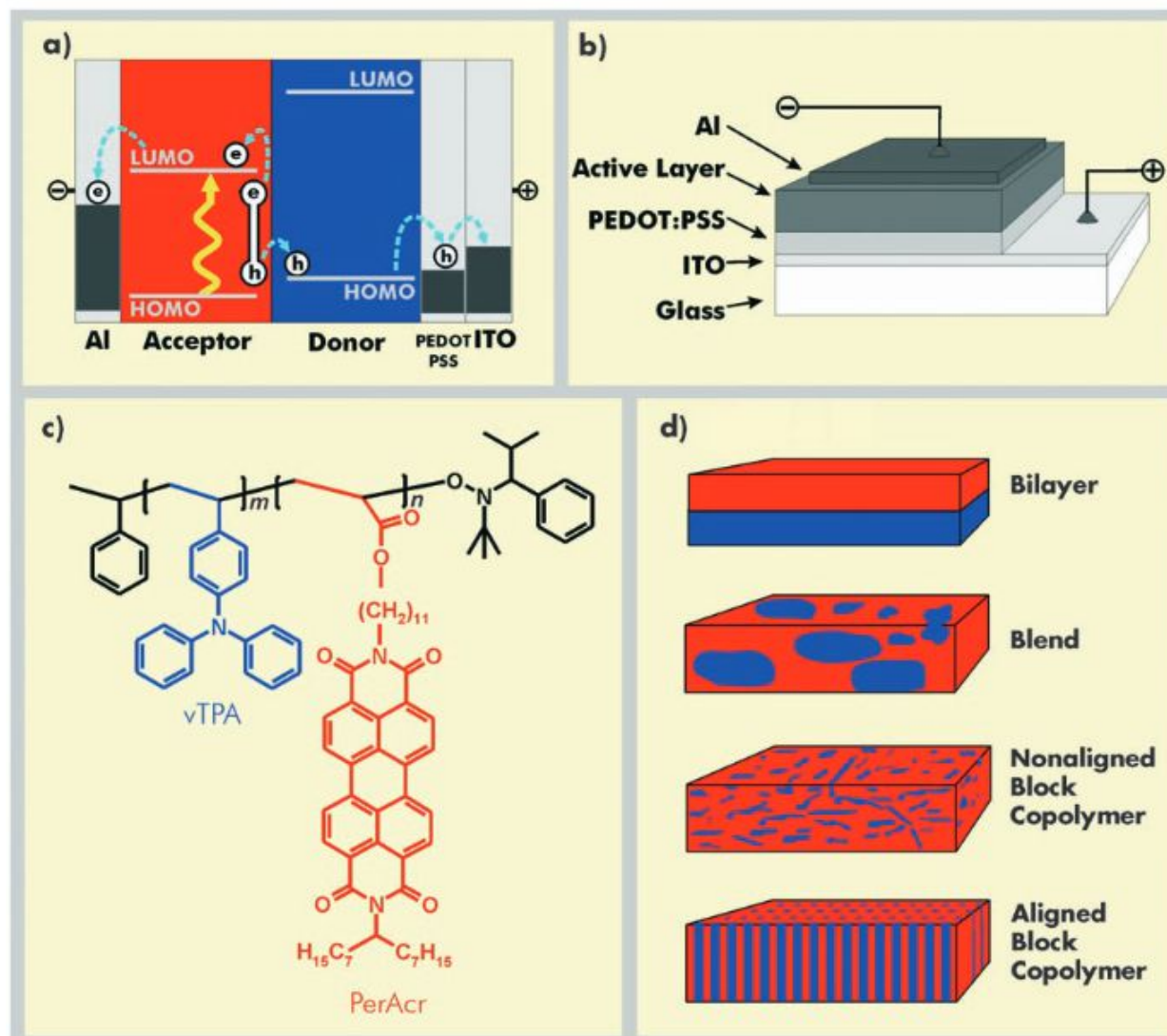


Applications of Bulk Self-Assembly

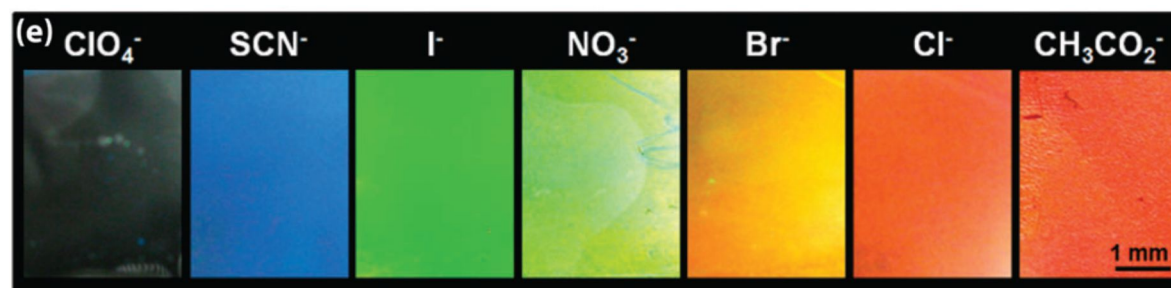
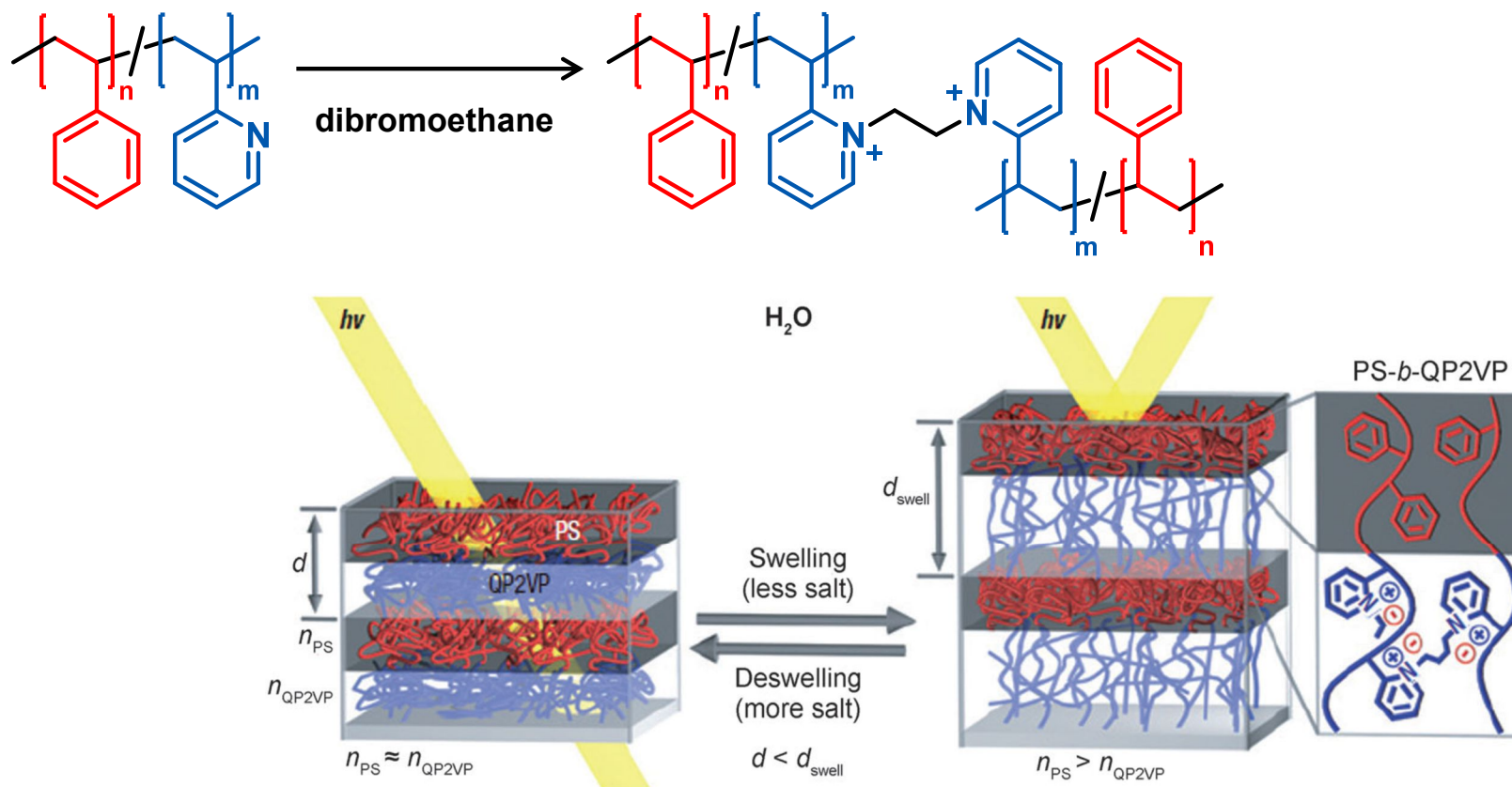
Selective Removal of One Domain

Templating for Materials Immobilization/Growth





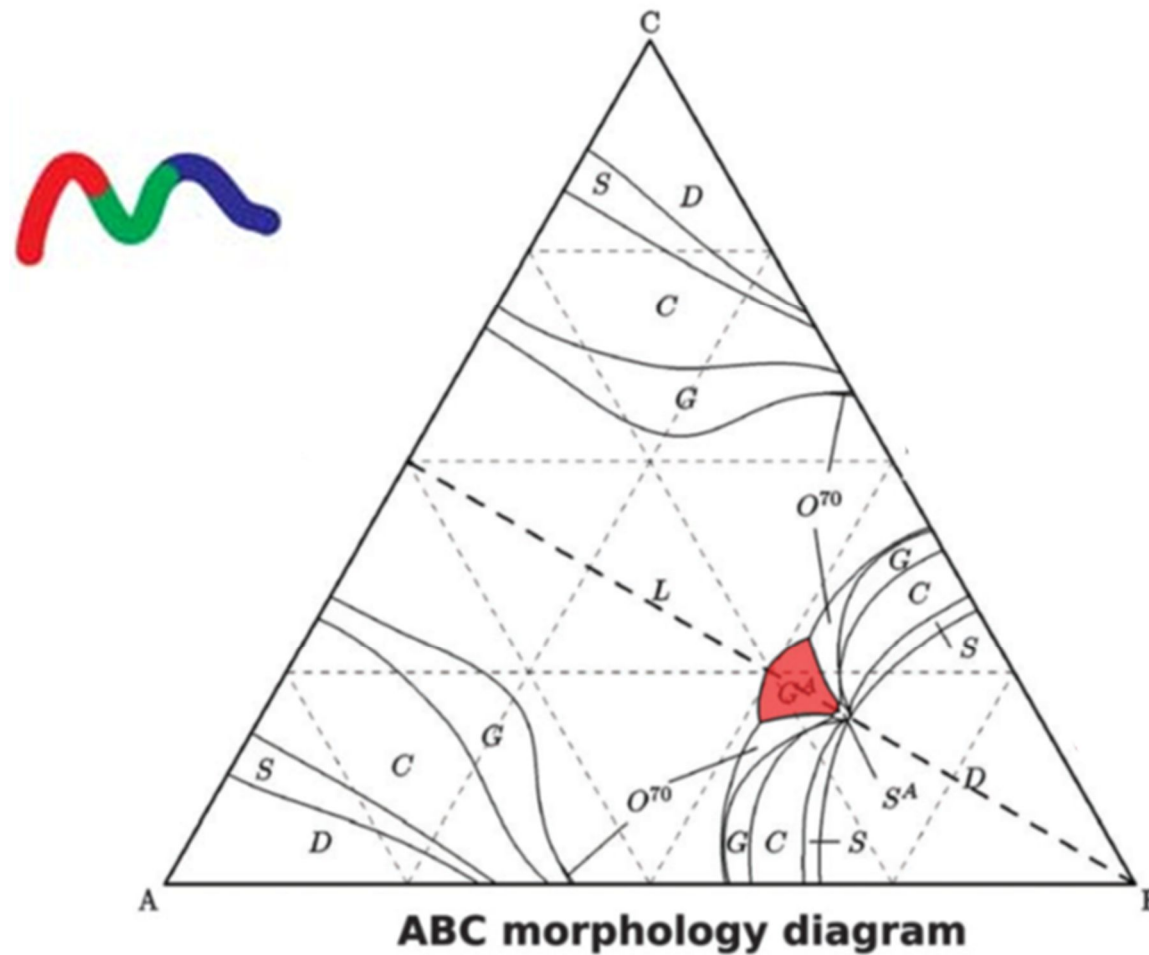
Photonic Materials



Nature Mater. 2007, 6, 957

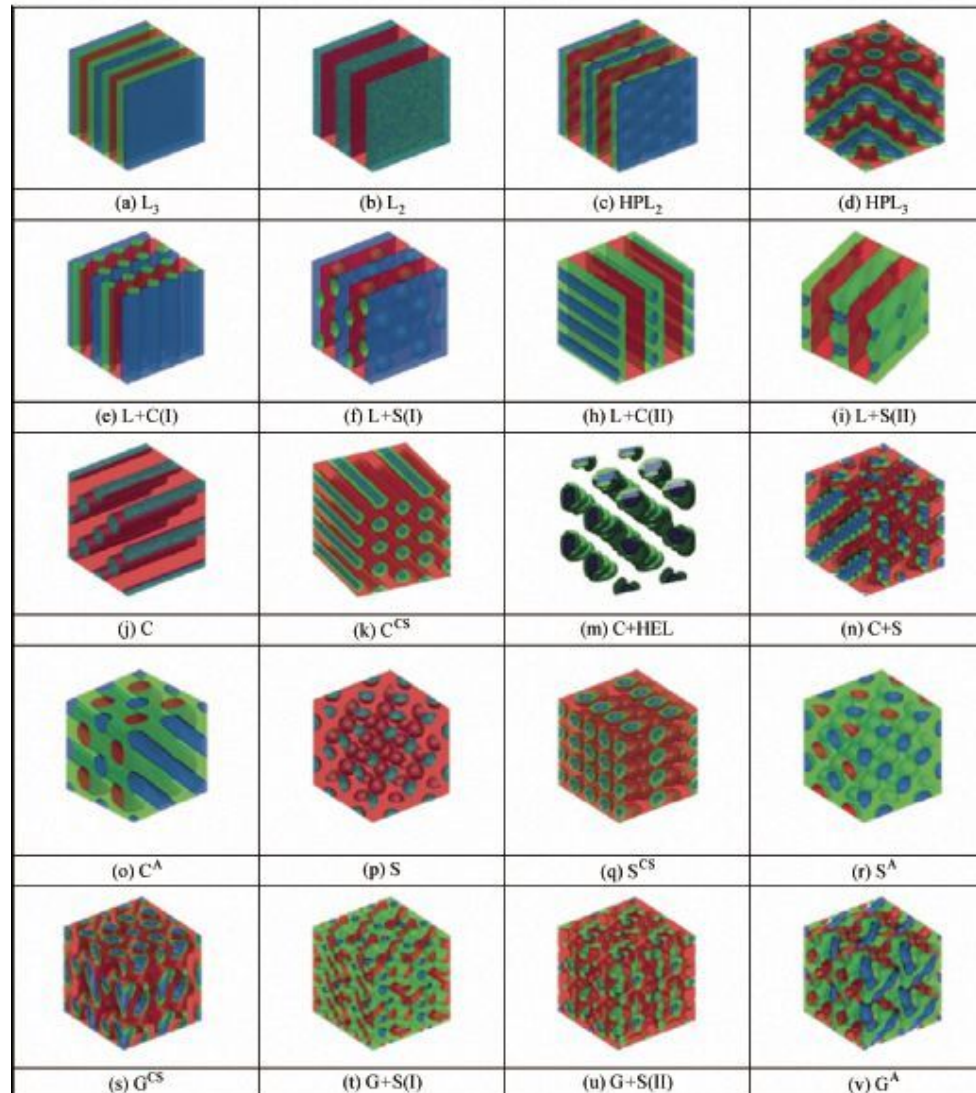
Photonic Materials

Selective Removal of One Domain – Triblock Copolymers



Photonic Materials

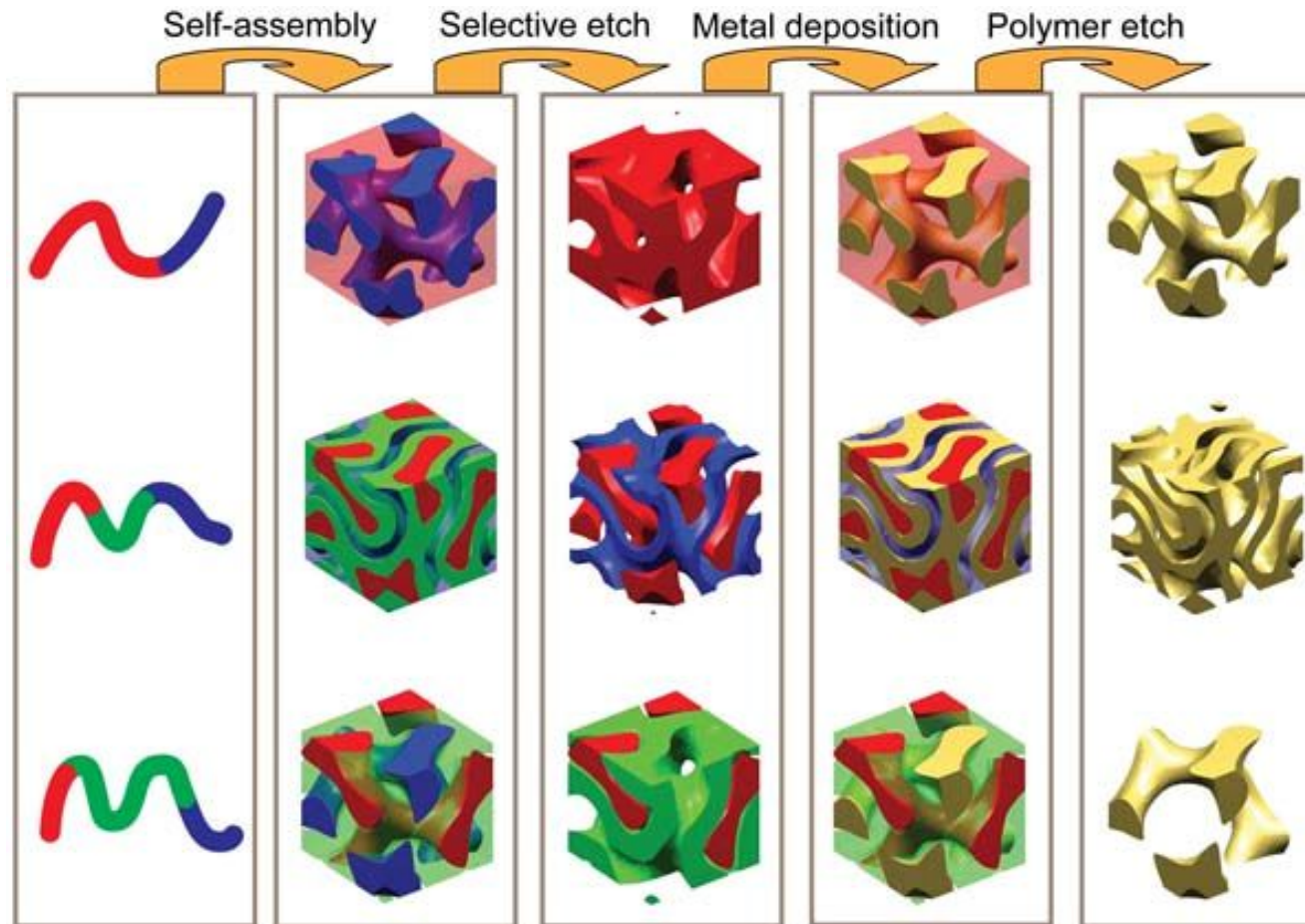
Selective Removal of One Domain – Triblock Copolymers



Applications of Bulk Self-Assembly

Photonic Materials

Selective Removal of One Domain – Triblock Copolymers



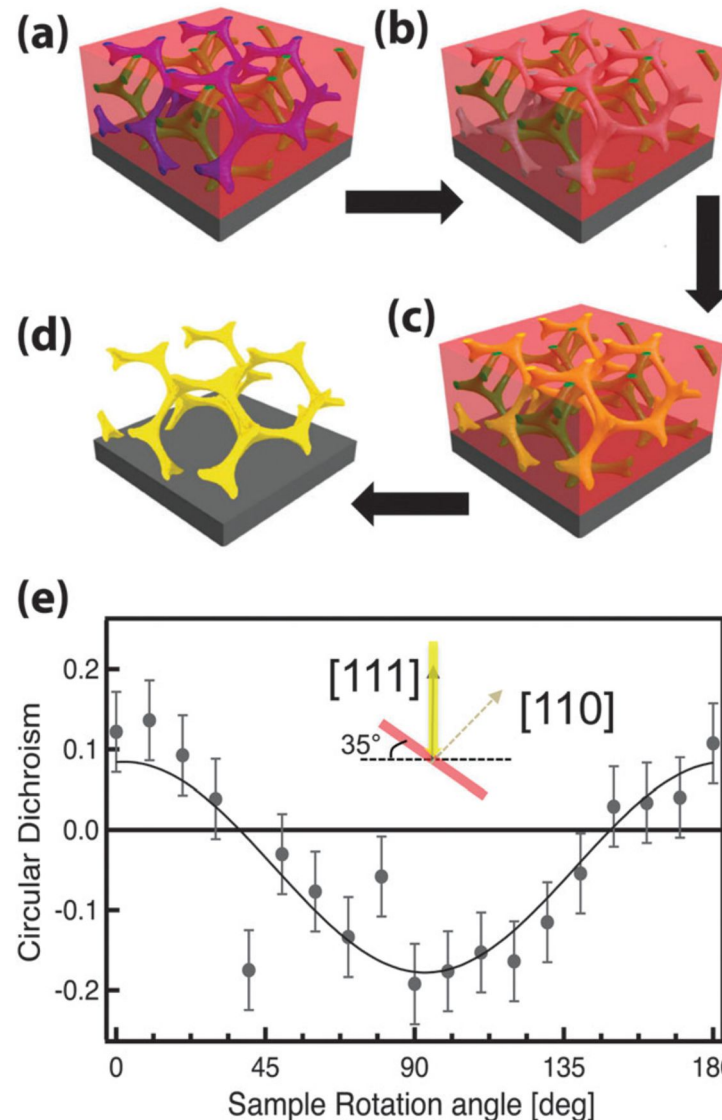
Process sometimes referred to as “hybridization”

Nature Mater. **2007**, 6, 957

Photonic Materials

Selective Removal of One Domain – Triblock Copolymers

PI-*b*-PS-*b*-PEO

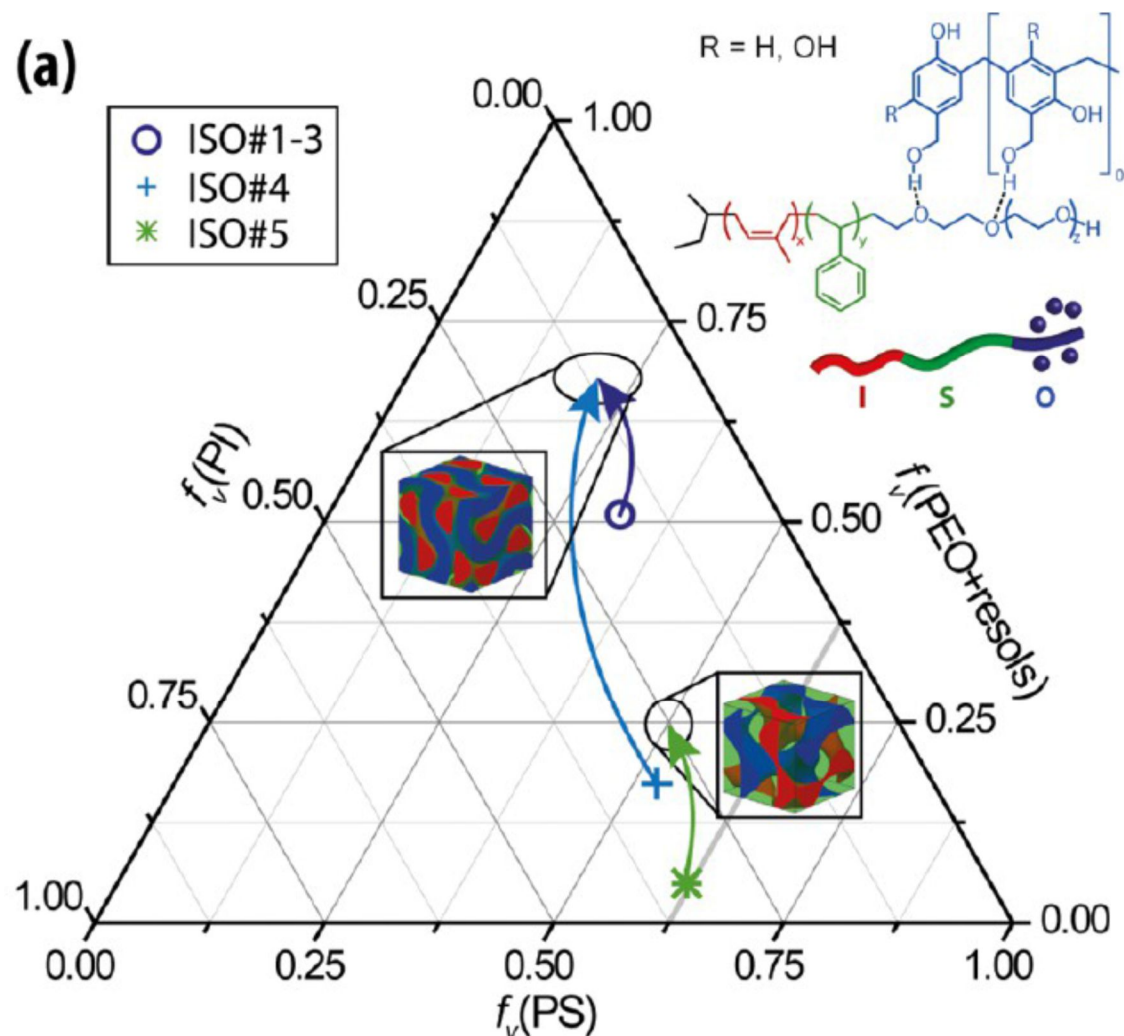


A **metamaterial** (from the Greek word μετά meta, meaning "beyond" and the Latin word materia, meaning "matter" or "material") is a material engineered to have a **property that is not found in naturally occurring materials**.

Ex.: negative-index metamaterials, cloaking

The resulting material exhibits preferential transmission of polarized light depending on its orientation.

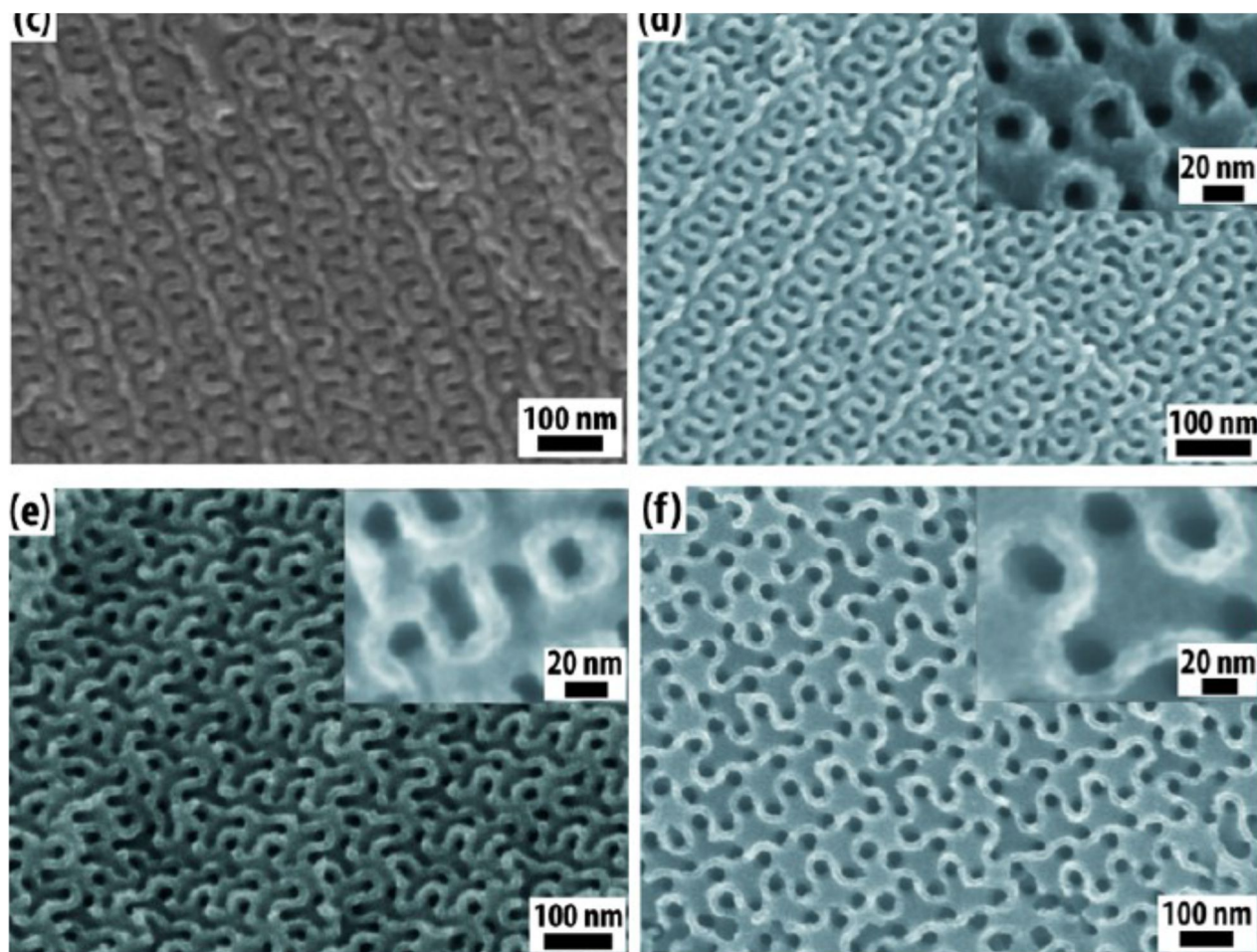
Selective Removal of One Domain Carbon Nanostructure Template



Applications of Bulk Self-Assembly

Selective Removal of One Domain

Carbon Nanostructure Template



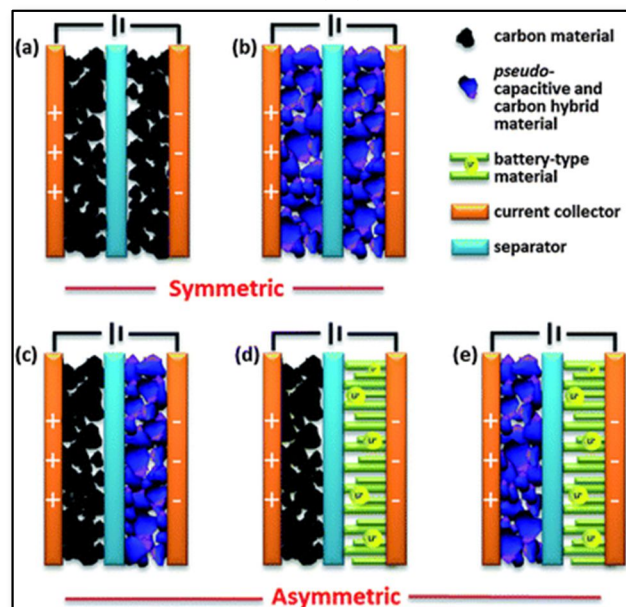
Selective Removal of One Domain

Carbon Nanostructure Template

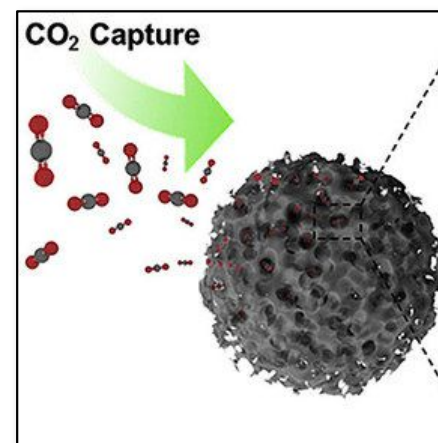
Meso/nanoporous carbon materials

Electrochemical capacitors (energy storage)

J. Mater. Chem. A **2013**, 1, 9395

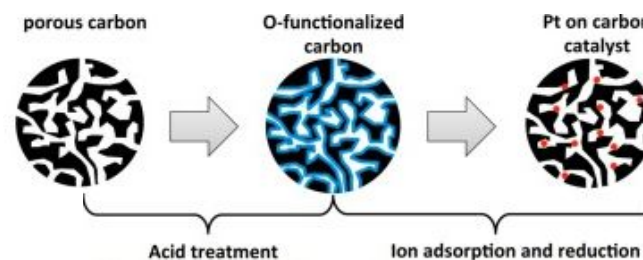


Pollutants absorbers



Carbon **2019**, 143, 531

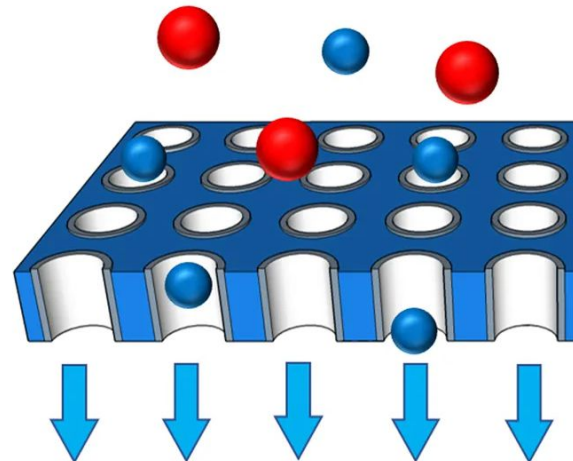
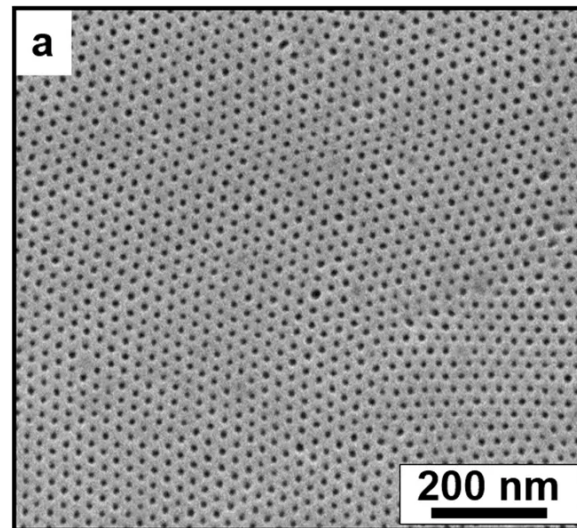
Catalysts



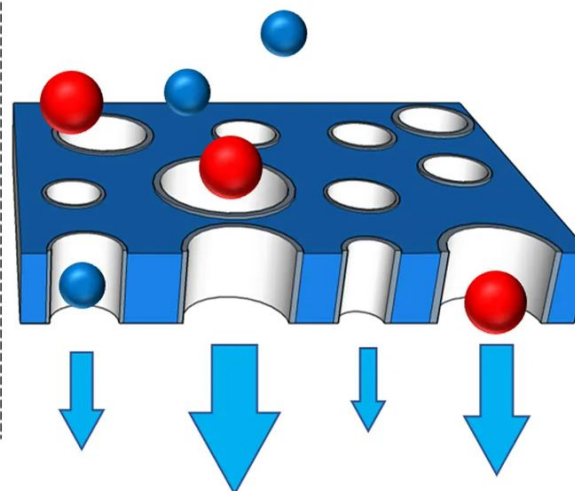
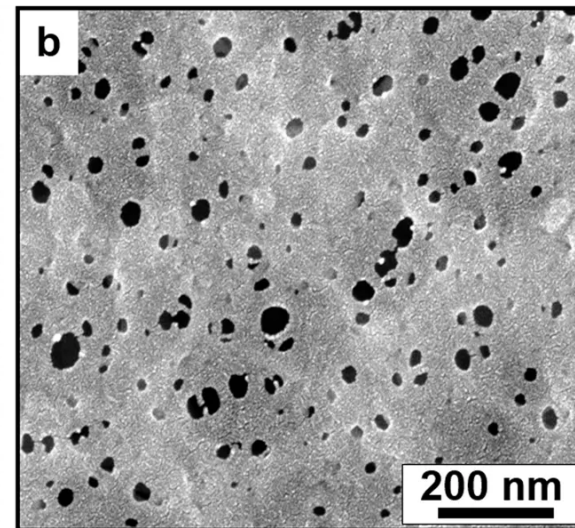
Catalysis Today **2015**, 249, 30

Selective Removal of One Domain Membranes

self-assembled
block copolymer membrane



commercial membrane made
using a standard phase
separation process

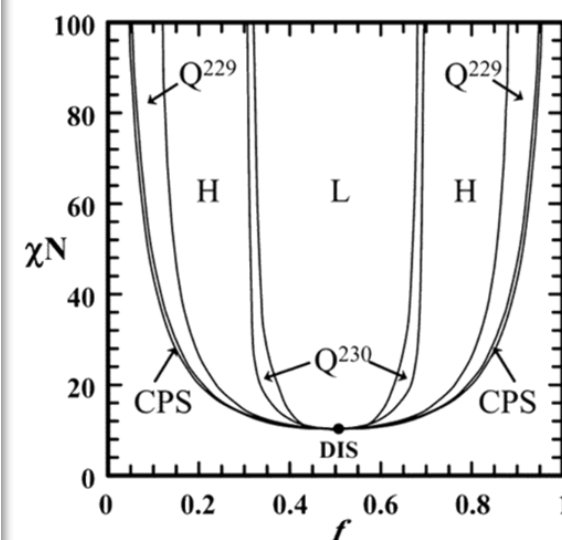


Selective Removal of One Domain Membranes

Table 1: Block copolymers (BCs), sacrificial components (SC), and the microstructures of the porous polymeric architectures.

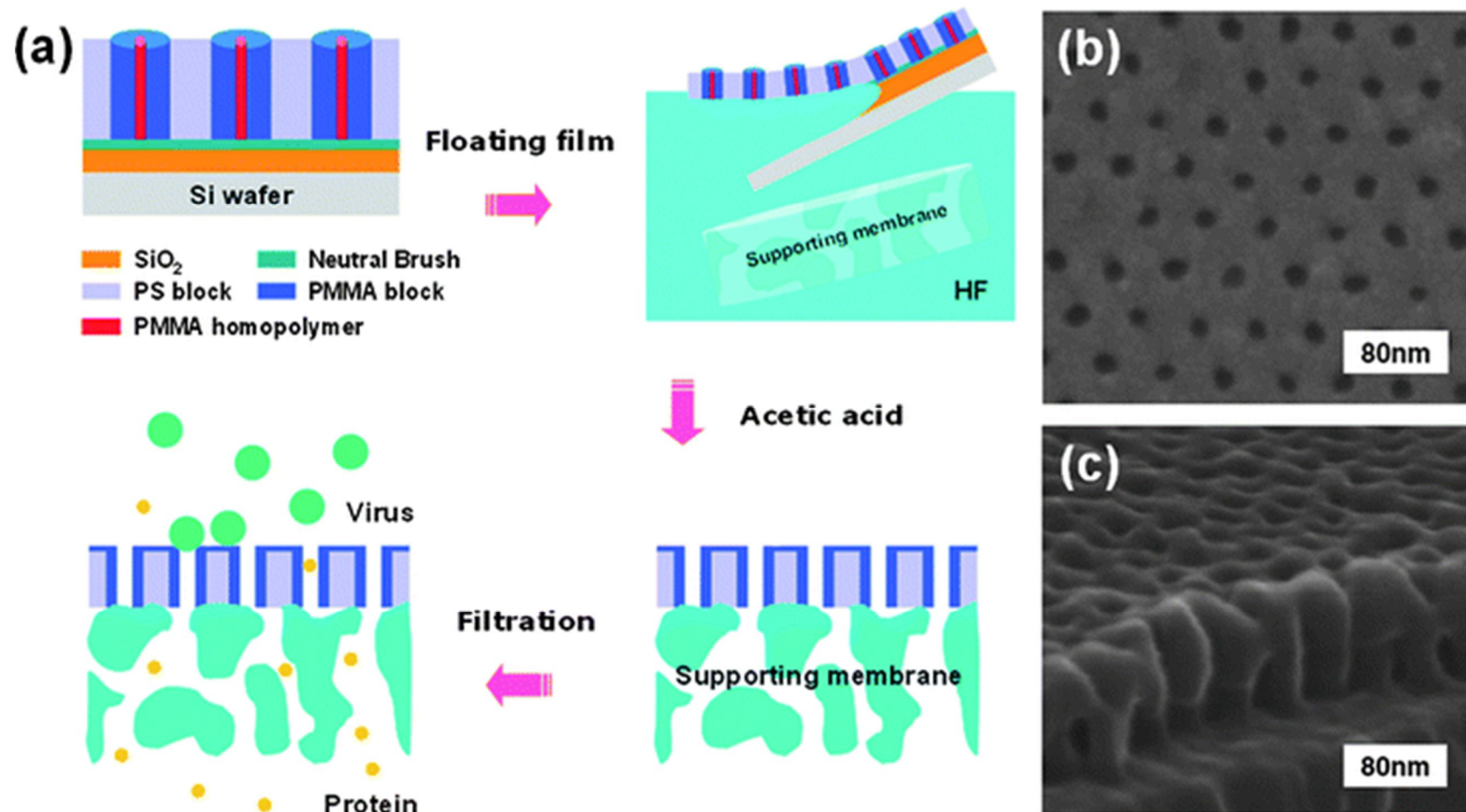
BCs ^a	SC	etching conditions	microstructure	ref.
1,2-PB- <i>b</i> -PDMS	PDMS	tetrabutylammonium fluoride in THF	H	[46]
P2VP- <i>b</i> -PI, PI h.	PI	ozonolysis	Q ²³⁰	[47,48]
P3DDT- <i>b</i> -PLA	PLA	NaOH	H	[49]
(P3HT-NH ₃ ⁺)- <i>b</i> -(PS-SO ₃ ⁻)	PS-SO ₃ ⁻	acetate, triethylamine	n.d.	[50]
PFS- <i>b</i> -PLA	PLA	NaOH	H, Q ²³⁰	[51,52]
PI- <i>b</i> -PS, PI h., PS h.	PS h.	hexane	Q ²³⁰	[53]
PFS- <i>b</i> -PMMA	PMMA	UV radiation	Q ²³⁰	[54]
PE- <i>b</i> -PEP, PE h., PEP h.	PEP h.	THF	Q ²³⁰	[55]
PE- <i>b</i> -PS	PS	fuming HNO ₃	H, Q ²³⁰	[56,57]
PLA- <i>b</i> -P(N-S)	PLA	NaOH	H	[58]
PS-<i>b</i>-PEO	PEO	HI or heating	H, Q ²³⁰	[59-63]
PS-<i>b</i>-PEO, resorcinol	resorcinol	2-propanol	H, Q ²³⁰	[45,64]
PS- <i>b</i> -PLA	PLA	NaOH or HI	H, Q ²³⁰	[65-69]
PS- <i>b</i> -PDMS	PS	O ₂ plasma treatment	H	[70]
PS- <i>b</i> -PMMA	PMMA	UV radiation	H, Q ²³⁰	[71-74]
PS- <i>b</i> -PMMA, PMMA h.	PMMA h.	CH ₃ COOH	H	[75,76]
PSTPA- <i>b</i> -PLA	PLA	NaOH	H	[77]
PS(BCB)- <i>b</i> -PMMA	PMMA	UV radiation	H	[78]
PS(BCB)- <i>b</i> -PLA	PLA	NaOH	H	[79]

^ah.: homopolymer; 1,2-PB: 1,2-poly(butadiene); P2VP: poly(2-vinylpyridine); P3DDT: poly(3-dodecylthiophene); P3HT-NH₃⁺: aniline chain-end-functionalized poly(3-hexylthiophene); PDMS: poly(dimethylsiloxane); PE: poly(ethylene); PEO: poly(ethylene oxide); PEP: poly(ethylene-*alt*-propylene); PFS: poly(ferrocenylsilane); PI: poly(isoprene); PLA: poly(lactic acid); PMMA: poly(methyl methacrylate); P(N-S): poly(norbornenylethylstyrene-*s*-styrene); PS: poly(styrene); PS(BCB): poly(styrene-*r*-benzocyclobutene); PS-SO₃⁻: sulfonic acid chain-end-functionalized poly(styrene); PSTPA: poly(styrene) containing triphenylamine side group.



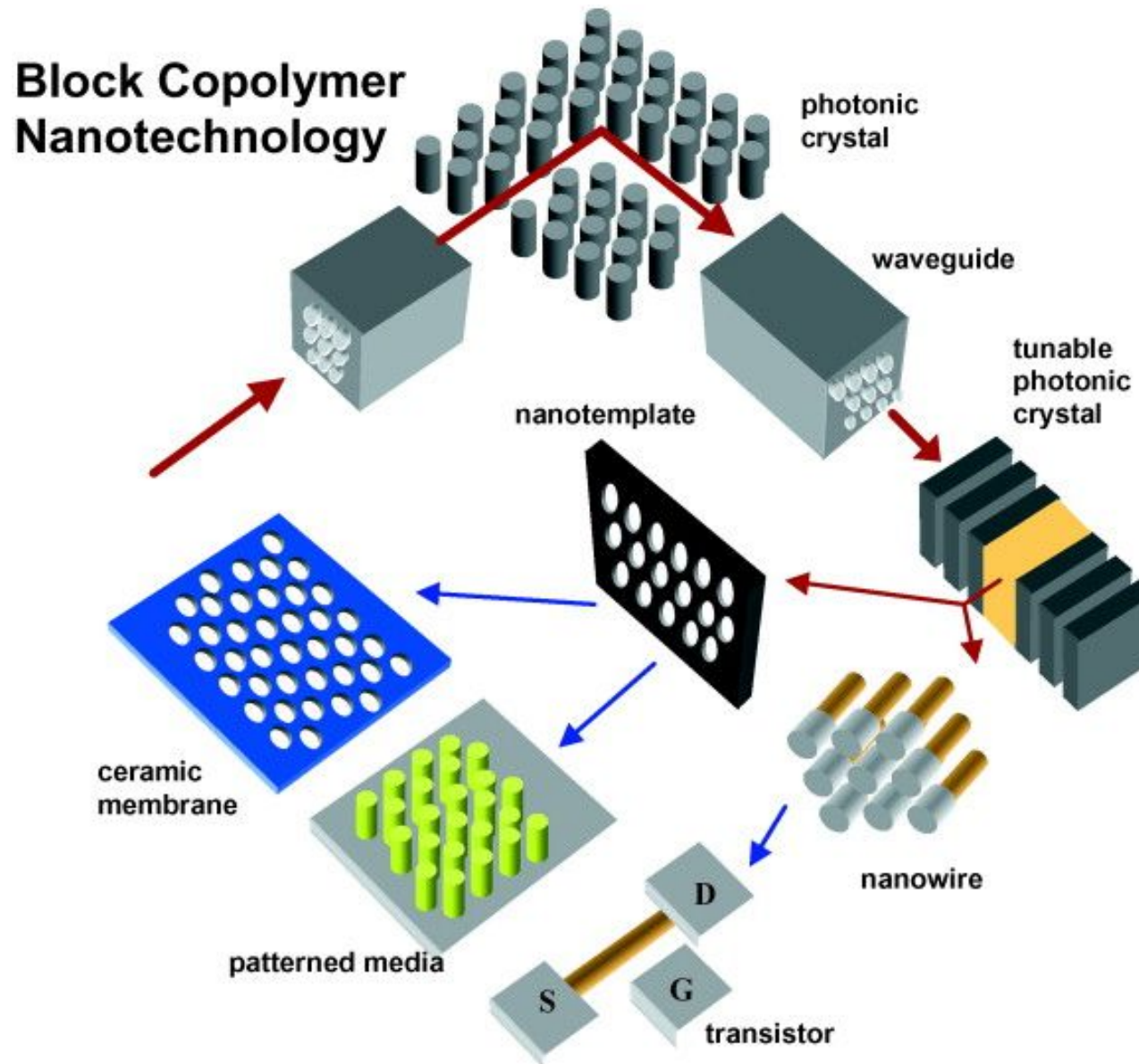
Applications of Bulk Self-Assembly

Selective Removal of One Domain Membranes



Applications of Bulk Self-Assembly

General Picture



Polymer **2003**, 44, 6725